# Order Processing & Workflow Architecture

This document explains how an order is punched, processed, routed, and completed using various modules in the WIN07 system.

## System Flowchart

```
CUSTOMER / RETAIL DESK
        |
        v
Retail Punching UI (retail_punching.py)
        |
        v
Cart Creation (Session State)
        |
        v
Order Pipeline (order_pipeline.py)
        |
        v
convert_cart_to_order (order_engine.py)
        |
        v
Validation Gateway (validation_gateway.py)
        |
        v
SQL Adapter → save_order()
        |
        v
PostgreSQL Database
        |
        v
Backoffice Dashboard
(backoffice_management.py)
        |
        v
Workflow Engine (workflow/engine.py)
        |
        v
Power Engine (power_engine.py)
        |
        v
Batch Manager (batch_manager.py)
        |
        v
Routing Engine
(INTERNAL / EXTERNAL / VENDOR)
        |
        v
Job Cards / Lab Orders / Labels
        |
        v
Billing & Status Update
        |
        v
Order Closure
```

## Module Responsibilities

1. retail_punching.py
   - Captures RX, product, quantity, patient data
   - Builds cart

2. order_pipeline.py
   - Converts cart to order
   - Triggers validation
   - Saves order

3. order_engine.py

- Builds standardized order structure

4. validation_gateway.py
   - Credit, fraud, pricing validation

5. sql_adapter.py
   - Database CRUD operations
   - fetch_orders, save_order, fetch_full_order

6. backoffice_management.py
   - Main workflow UI
   - Allocation, pricing, job cards

7. power_engine.py
   - Vertex correction
   - Manufacturing power

8. batch_manager.py
   - FIFO allocation
   - Stock lookup

9. workflow/engine.py
   - Orchestrates complete workflow
   - Routing + status

10. workflow/status.py
   - Order status lifecycle

## Data Flow Summary

```
Retail UI → Session → Order Dict → Database → Backoffice UI
          → Workflow Engine → Production / Vendor → Billing
```

## Status Lifecycle

```
PENDING
   ↓
CONFIRMED
   ↓
ALLOCATED
   ↓
IN_PRODUCTION
   ↓
READY
   ↓
DISPATCHED
   ↓
DELIVERED
```

## Future Expansion Plan

- Vendor API integration
- Auto PO generation
- Lab SLA monitoring
- Real-time stock sync
- Analytics dashboard

## Conclusion

This workflow ensures complete traceability from punching to delivery. The modular architecture allows independent scaling of retail, lab, and vendor operations.